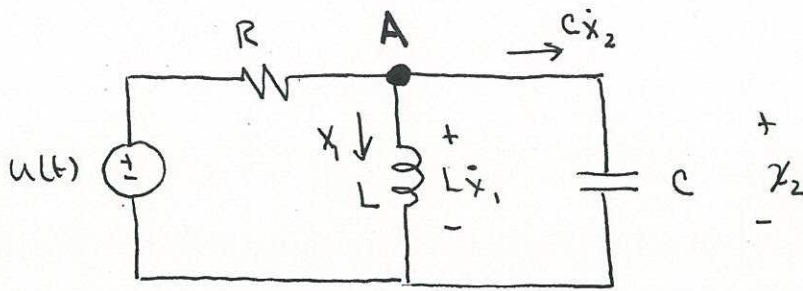


9.1



- (1) Define state variables $x_1(t)$ & $x_2(t)$ as shown above
- (2) Label voltage across inductor ($L\dot{x}_1$) and current through capacitor ($C\dot{x}_2$)

$$\text{KCL at A: } \frac{u - x_2}{R} = \dot{x}_1 + C\dot{x}_2 \Rightarrow \underline{\dot{x}_2 = \frac{u}{RC} - \frac{1}{RC}x_2 - \frac{1}{C}x_1}$$

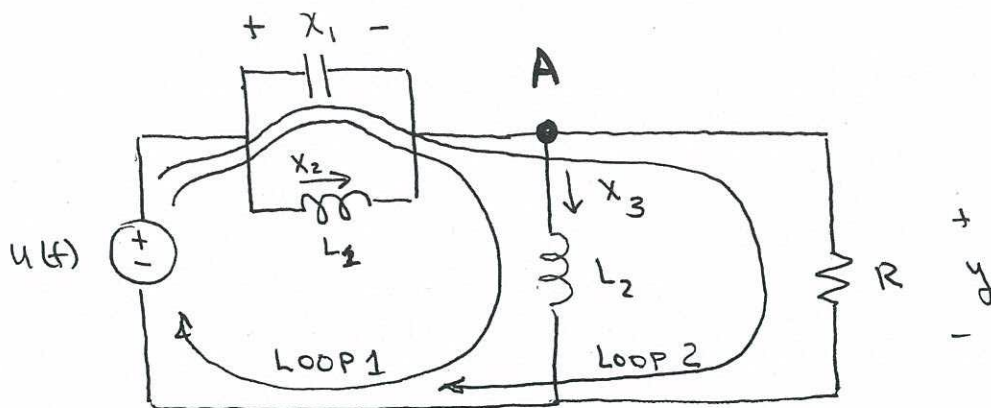
$$\text{KVL, right most loop: } L\dot{x}_1 = x_2 \Rightarrow \underline{\dot{x}_1 = \frac{1}{L}x_2}$$

$$\text{output equation: } y = x_2$$

Matrix form:

$$A = \begin{bmatrix} 0 & \frac{1}{L} \\ -\frac{1}{C} & -\frac{1}{RC} \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} 0 \\ \frac{1}{RC} \end{bmatrix}, \quad \underline{c} = [0 \quad 1], \quad d=0$$

9.2



$$\text{KVL, Loop 1: } u = x_1 + L_2 \dot{x}_3 \Rightarrow \underline{\dot{x}_3 = -\frac{1}{L_2} x_1 + \frac{1}{L_2} u}$$

$$\text{KCL at A: } x_2 + C \dot{x}_1 = x_3 + \frac{y}{R}$$

$$\text{KVL, Loop 2: } u - x_1 - y = 0 \Rightarrow y = u - x_1$$

$$x_2 + C \dot{x}_1 = x_3 + \frac{u}{R} - \frac{x_1}{R}$$

$$\underline{\dot{x}_1 = -\frac{x_1}{RC} - \frac{x_2}{C} + \frac{x_3}{C} + \frac{u}{RC}}$$

Parallel inductor, L_1 , and capacitor:

$$x_1 = L_1 \dot{x}_2 \Rightarrow \underline{\dot{x}_2 = \frac{1}{L_1} x_1}$$

output equation (from KVL, Loop 2 above):

$$\underline{y = u - x_1}$$

↓ continued

9.2 (cont'd)

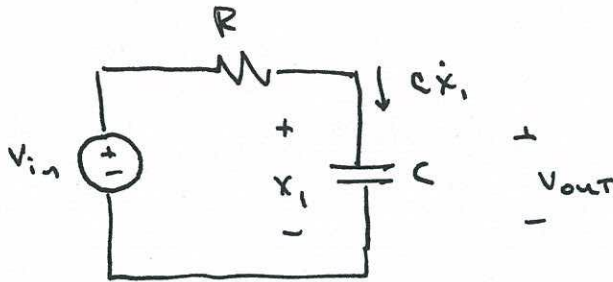
In matrix form:

$$A = \begin{bmatrix} -\frac{1}{RC} & -\frac{1}{C} & \frac{1}{C} \\ \frac{1}{L_1} & 0 & 0 \\ -\frac{1}{L_2} & 0 & 0 \end{bmatrix}, \quad \underline{b} = \begin{bmatrix} \frac{1}{RC} \\ 0 \\ \frac{1}{L_2} \end{bmatrix}$$

$$\underline{c} = [-1 \quad 0 \quad 0], \quad d = 1$$

9.3

(a)



$$\text{KVL: } V_{in}(t) = RC\dot{x}_1 + x_1 \Rightarrow \underline{\underline{\dot{x}_1 = -\frac{1}{RC}x_1 + \frac{1}{RC}V_{in}}}$$

$$\text{Output equation: } V_{out} = x_1$$

$$A = \left[-\frac{1}{RC} \right], \quad b = \frac{1}{RC}, \quad c = 1, \quad d = 0$$

(b) Matlab simulations with, $R = 470\Omega$, $C = 1\mu F$, $V_{in} = 4u_0(t)$:

Matlab commands:

```
R = 470;
```

```
C = 1e-6;
```

```
A = [-1/R/C];
```

```
b = [1/(R*C)];
```

```
c = 4*[1]; % scale output to correspond to input = 4*u_0(t)
```

```
d = 0;
```

```
sys = ss(A,b,c,d);
```

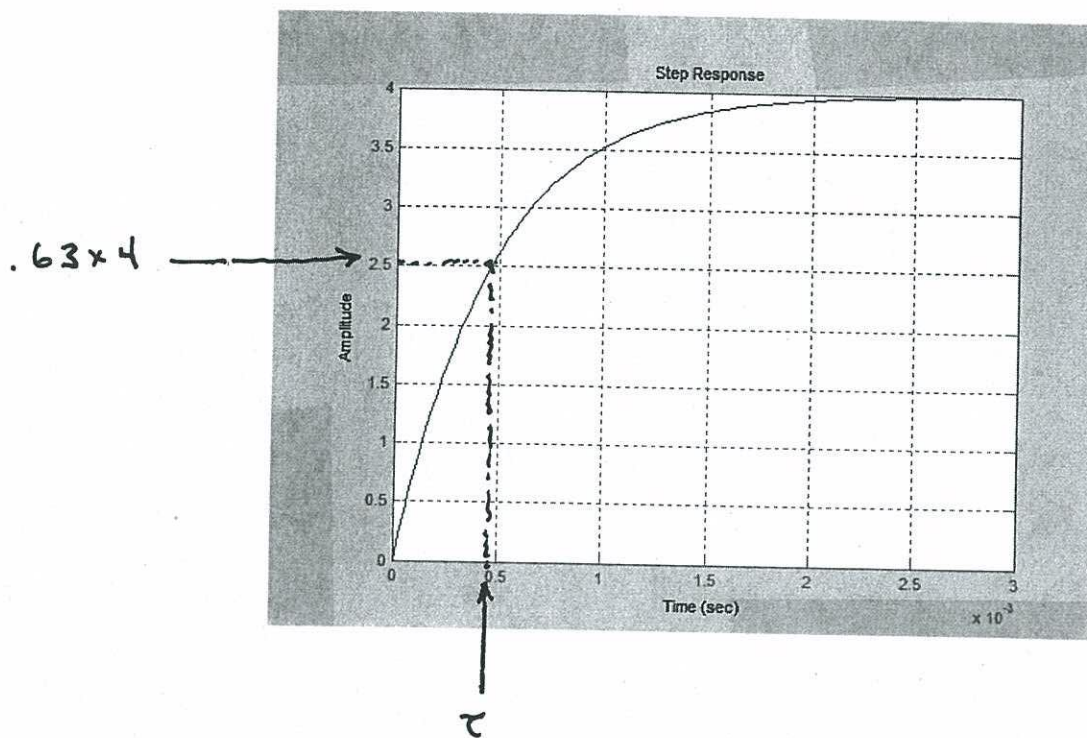
```
step(sys)
```

```
grid
```

{ continued

9.3 (cont'd)

(b) Simulated response (from MATLAB):



(c) From circuit :

$$\tau = R_{eq} C = (470 \Omega)(1 \times 10^{-6} \text{ F}) \approx 4.7 \text{ msec}$$

From plot above, $\tau \approx 4.8 \text{ msec}$

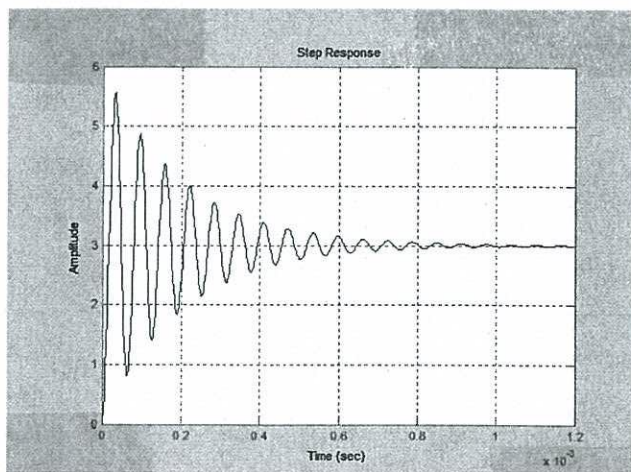
Agreement to within
the accuracy to which
I can read the graph

9.4 The state variable model for this problem was created in exercise 1 of chapter 9.1. We just need to plug in numbers and do the simulation.

Matlab commands:

```
R = 10;  
L = 1e-3;  
C = 100e-9;  
  
A = [-R/L -1/L; 1/C 0];  
b = [1/L; 0];  
c = 3*[0 1]; % scale output to account for Vin = 3*u0(t)  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Resulting Figure:



↖ The scale on the default figure is not too useful for reading rise time and peak value.

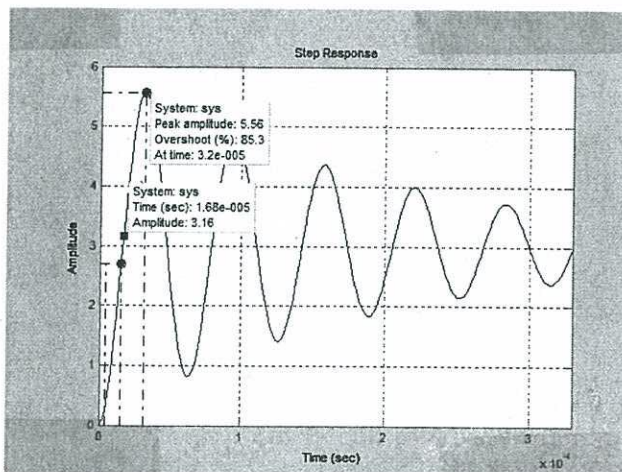
We can, however, modify it...

↓ (cont'd)

9.4 (cont'd)

- MATLAB makes it easy to zoom in on a portion of the figure. click on the magnifying glass on the toolbar, and use your mouse to select the section of the figure to enlarge.
- If the response was generated using MATLAB's "step" command, MATLAB will display certain parameters on the plot. Right-click on the figure, click on "characteristics" and select the value you want displayed.

I labeled the figure as shown below:



$\Leftarrow$ peak value = 5.56
 $t_r = 1.68 \times 10^{-5} \text{ sec}$

↓ (cont'd)

9.4 (st.11 continued)

(c) The input-output relation is:

$$\frac{d^2 V_{out}}{dt^2} + \frac{R}{L} \frac{dV_{out}}{dt} + \frac{1}{LC} V_{out} = \frac{1}{LC} V_{in}$$

$$\omega_n^2 = \frac{1}{LC} \Rightarrow \omega_n = \sqrt{\frac{1}{(.001)(100 \times 10^{-9})}} = 100,000 \text{ rad/s}$$

$$2\zeta\omega_n = \frac{R}{L} \Rightarrow \zeta = \frac{10}{2(.001)(100,000)} = 0.05$$

$$M_p \approx 1 - \frac{\zeta}{0.6} = 0.91$$

$$y_p = y_{ss} + y_{ss} \cdot M_p = 3V + (3V)(0.91) = \underline{5.73V}$$

$$t_r \approx \frac{1.8}{\omega_n} = \underline{1.8 \times 10^{-5} \text{ sec}}$$

These values are very close to the simulation values, considering the approximations used!

9.5 The state variable model for this circuit was created in exercise 2 of chapter 9.2. We just need to plug in numbers and run the simulation.

(b)

Matlab commands:

```
R1 = 10;  
R2 = 100;  
R3 = 1.1;  
L = 1e-3;  
C = 10e-6;  
  
A = [-1/R3/L 1/L; -1/C -1/C*(1/R1+1/R2)];  
b = [0; 1/(R1*C)];  
c = 2*[0 1]; % Scale to account for input = 2*u0(t)  
d = 0;  
  
sys = ss(A,b,c,d);  
step(sys)
```

Figure:

